

GOVERNMENT OF INDIA - MINISTRY OF EARTH SCIENCES / IMD

MausamMesh (■■■■■■■■■) - Panchayat Weather Intelligence Bulletin

Hyperlocal Weather Intelligence for Precision Farming (IMD & MoES Prototype Emblem Placeholder)

Location: India > Maharashtra > Pune > Haveli > Wagholi	Generated At: 2026-09-28 01:52 IST
Temp: 28.5°C Humidity: 88.0%	Model Version: v3.0-MinT-Conformal-XGBoost
Skill Gate: Passed Skill Gate (Model beats Block Baseline)	Telemetry: Active Telemetry Connector

1. 5-Day Downscaled Weather Forecast Summary

Lead Day	Date & Day	Block Base (mm)	Panchayat Rain (mm)	90% Conformal Range	Risk Level	Expected Condition
Day 1	Mon 2026-09-28	24.5 mm	24.5 mm	24.0 - 25.1 mm	High	Heavy Rain Spells
Day 2	Tue 2026-09-29	18.0 mm	18.5 mm	17.9 - 19.1 mm	High	Moderate Rain
Day 3	Wed 2026-09-30	8.5 mm	9.0 mm	8.4 - 9.5 mm	Moderate	Light Passing Showers
Day 4	Thu 2026-10-01	2.0 mm	2.1 mm	1.5 - 2.7 mm	Low	Partly Cloudy
Day 5	Fri 2026-10-02	14.2 mm	15.2 mm	14.6 - 15.8 mm	High	Scattered Thunderstorms

2. Model Technical Verification & Skill Metrics

- Overall Model MAE: 0.178 mm vs Block Baseline MAE: 0.331 mm (+46.2% MAE Reduction)
- RMSE: 0.418 mm | Critical Success Index (CSI): 0.92 | Coverage Guarantee: 87.4%
- Hierarchical Coherence (MinT): Panchayat forecasts mathematically aggregate to parent Block forecast average.

3. Agro-Meteorological Crop Advisory (Cotton)

- Target Crop: Cotton
- Irrigation Management: Postpone irrigation for the next 48 hours due to expected rainfall spells (24.5 mm). Ensure surface drainage in low-lying plots.
 - Plant Protection: High humidity (88.0%) and warm temperatures favor sucking pest and fungal spore incidence. Spray recommended biopesticides during dry morning windows.
 - DAMU Nodal Approval: Advisory reviewed & verified by AMFU Nodal Officer (Status: APPROVED).

Disclaimer: Prototype Agro-Meteorological Advisory generated by SIH26074 Downscaling Engine. Data sources: WINDS ARG Telemetry & IMD Regional Forecasts. For official operational decisions, refer to District AMFU bulletins.